

# VRNQ — VR SOFTWARE RATING SHEET

Virtual Reality Neuroscience Questionnaire (Kourtesis et al., 2019) — one sheet per game, per rater.

Completed by trained adult raters (expert panel / facilitators) after hands-on use of the game. Not for completion by child participants.

**BEFORE USE** — The 20 item wordings must be transcribed verbatim from the source paper into the rows below. The grey labels name each item's *content*, not the published wording. Do not administer this sheet until the real item text is in place.

| Game rated | Rater ID | Role | Date | Minutes in this game | Input (controller / gaze) |
|------------|----------|------|------|----------------------|---------------------------|
|------------|----------|------|------|----------------------|---------------------------|

| Rate each item 1–7                     | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|----------------------------------------|---|---|---|---|---|---|---|
| 1 = extremely low · 7 = extremely high |   |   |   |   |   |   |   |

## A. User experience

*intensity of immersion, level of enjoyment, quality of graphics, quality of sound, quality of the VR technology (internal and external hardware)*

|                                                                          |   |   |   |   |            |   |   |
|--------------------------------------------------------------------------|---|---|---|---|------------|---|---|
| 1. [transcribe VRNQ item wording]<br><i>content: immersion</i>           | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 2. [transcribe VRNQ item wording]<br><i>content: enjoyment</i>           | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 3. [transcribe VRNQ item wording]<br><i>content: graphics quality</i>    | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 4. [transcribe VRNQ item wording]<br><i>content: sound quality</i>       | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 5. [transcribe VRNQ item wording]<br><i>content: VR hardware quality</i> | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| <b>A sub-score</b>                                                       |   |   |   |   | range 5–35 |   |   |

## B. Game mechanics

*ease of navigating, moving physically, and interacting with the virtual environment (using, picking and placing, holding items; two-handed interactions)*

|                                                                                              |   |   |   |   |            |   |   |
|----------------------------------------------------------------------------------------------|---|---|---|---|------------|---|---|
| 6. [transcribe VRNQ item wording]<br><i>content: ease of navigation</i>                      | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 7. [transcribe VRNQ item wording]<br><i>content: physical movement</i>                       | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 8. [transcribe VRNQ item wording]<br><i>content: interaction / using items</i>               | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 9. [transcribe VRNQ item wording]<br><i>content: pick and place items</i>                    | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 10. [transcribe VRNQ item wording]<br><i>content: holding items / two-handed interaction</i> | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| <b>B sub-score</b>                                                                           |   |   |   |   | range 5–35 |   |   |

## C. In-game assistance

*quality and helpfulness of the in-game instructions and prompts*

|                                                                            |   |   |   |   |            |   |   |
|----------------------------------------------------------------------------|---|---|---|---|------------|---|---|
| 11. [transcribe VRNQ item wording]<br><i>content: in-game instructions</i> | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 12. [transcribe VRNQ item wording]<br><i>content: in-game prompts</i>      | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 13. [transcribe VRNQ item wording]<br><i>content: see source</i>           | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 14. [transcribe VRNQ item wording]<br><i>content: see source</i>           | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 15. [transcribe VRNQ item wording]<br><i>content: see source</i>           | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| <b>C sub-score</b>                                                         |   |   |   |   | range 5–35 |   |   |

## D. VRISE (VR-induced symptoms and effects)

*nausea, disorientation, dizziness, fatigue, instability — CHECK ITEM DIRECTION IN THE SOURCE BEFORE SUMMING (see note on page 2)*

|                                                                      |   |   |   |   |            |   |   |
|----------------------------------------------------------------------|---|---|---|---|------------|---|---|
| 16. [transcribe VRNQ item wording]<br><i>content: nausea</i>         | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 17. [transcribe VRNQ item wording]<br><i>content: disorientation</i> | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 18. [transcribe VRNQ item wording]<br><i>content: dizziness</i>      | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 19. [transcribe VRNQ item wording]<br><i>content: fatigue</i>        | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| 20. [transcribe VRNQ item wording]<br><i>content: instability</i>    | 1 | 2 | 3 | 4 | 5          | 6 | 7 |
| <b>D sub-score</b>                                                   |   |   |   |   | range 5–35 |   |   |

|                                               |  |  |  |  |              |  |  |
|-----------------------------------------------|--|--|--|--|--------------|--|--|
| <b>TOTAL VRNQ SCORE (sum of all 20 items)</b> |  |  |  |  | range 20–140 |  |  |
|-----------------------------------------------|--|--|--|--|--------------|--|--|

## Scoring

- **Sub-score.** Sum the five items in each domain. Range 5–35.
- **Total.** Sum all twenty items. Range 20–140.
- **Cut-offs apply to the median across raters, not to one rater's sheet.** Collect sheets from all raters for a game, take the median sub-score and median total, then compare to the table below.

|                                    | Minimum cut-off<br>adequate quality | Parsimonious cut-off<br>robustly suitable |
|------------------------------------|-------------------------------------|-------------------------------------------|
| Each sub-score (range 5–35)        | <b>25</b>                           | <b>30</b>                                 |
| Total score (range 20–140)         | <b>100</b>                          | <b>120</b>                                |
| Equivalent to a per-item median of | 5                                   | 6                                         |

Reporting: state, per game, whether the median sub-scores and total met the minimum cut-offs, and whether they met the parsimonious cut-offs. A game meeting the minimum cut-offs is of adequate quality not to cause significant VRISE; meeting the parsimonious cut-offs supports suitability more robustly.

## Summary across the six games

Median scores across all raters. Mark a cell that falls below the minimum cut-off.

| Game               | A UX | B Mechanics | C Assistance | D VRISE | Total |
|--------------------|------|-------------|--------------|---------|-------|
| Emotion Room 360   |      |             |              |         |       |
| Emotion Cinema 360 |      |             |              |         |       |
| Playroom 360       |      |             |              |         |       |
| Football 360       |      |             |              |         |       |
| Museum 360         |      |             |              |         |       |
| Park 360           |      |             |              |         |       |

## Before you use this sheet

- **Transcribe the 20 items verbatim.** The rows on page 1 carry content labels, not the published item wording. Copy the real wording from the source paper before any rater sees the sheet.
- **Check the VRISE item direction.** Confirm in the source whether the VRISE items are worded so that a high score means fewer symptoms, and whether any item is reverse-scored, before you sum domain D.
- **Keep it in English.** There is no validated Malayalam version. If raters need Malayalam, commission a forward–back translation and report it as an adapted version, exactly as the study blueprint requires for the other instruments.
- **Raters are adults.** The VRNQ was validated on 40 adults aged 28–43. It rates the software, not the child. Child-level tolerance and enjoyment belong on the VR session user-experience form, not here.
- **Use more than one rater per game.** Report the number of raters and the spread of scores alongside the median. Cut-offs on a single rater's sheet mean little.

## Source

Kourtesis, P., Collina, S., Dumas, L. A. A., & MacPherson, S. E. (2019). Validation of the Virtual Reality Neuroscience Questionnaire: Maximum duration of immersive virtual reality sessions without the presence of pertinent adverse symptomatology. *Frontiers in Human Neuroscience*, 13, 417.

Verify the item wording, the scoring rules and the cut-off values against this paper before use. Check and follow its licence terms when reproducing the items.